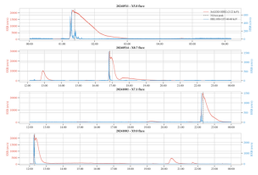
PHYSICS-INFORMED SPATIO-TEMPORAL GRAPH TRANSFORMER FOR SOLAR FLARE EARLY WARNING

B.Tech P2BL Project | AI & ML Specialization | Department of Computer Science & Engineering | Sathyabama Institute of Science and Technology

🚨 AI & ML Problem Statement

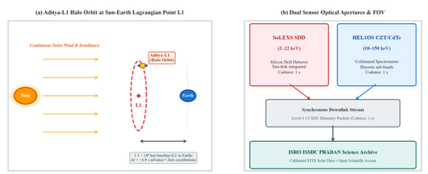
- **Rare-Event Imbalance (1:300)**: Major X-flares comprise <0.3% of solar hours; standard ML collapses to all-quiet majority class.
- **Cross-Sensor Temporal Delay**: Fast non-thermal electrons drive delayed plasma heating; black-box ML violates physical causality.



Aditya-L1 Telemetry: Non-Thermal HXR Burst Precedes Thermal SXR Peak (Neupert Effect)

💡 Proposed AI & ML Solution

- **5-Node Graph Transformer**: Models cross-spectral electromagnetic coupling across SoLEXS & HEL1OS sensors.
- **Differentiable PINN Regularization**: Embeds Neupert plasma equation with GradNorm multi-task dynamic balancing.



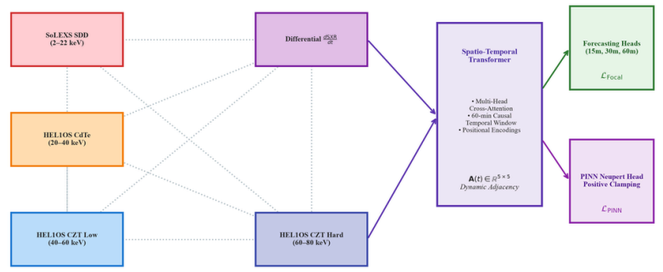
Aditya-L1 Orbit at Sun-Earth L1 (1.5M km) with Dual X-Ray Spectrometers

⚙️ Key Machine Learning Innovations

- **Spatial & Causal Attention**: Dual-axis multi-head attention over passbands and sliding time windows.
- **Positive Parameter Clamping**: Learnable coronal cooling (β ≥ 0.001) & non-thermal injection (α ≥ 0.01).
- **Mondrian Conformal Sets**: 95% guaranteed coverage prediction bounds eliminating false alarms.

🧠 Spatio-Temporal Graph Transformer Neural Architecture

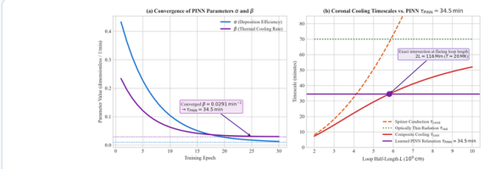
PyTorch Deep Learning Model



Multi-Task Thermodynamic PINN Loss:

```
L_total = L_focal + λ(t) × || d(SXR)/dt - (α × HXR - β × SXR) ||²
```

- Dynamically balanced via **GradNorm** (α ≥ 0.01, β ≥ 0.001)
- Constrains predictions to adhere to plasma energy conservation.



PINN Cooling Parameter Convergence vs. Spitzer Loss

🔄 End-to-End AI/ML Training & Real-Time Operational Pipeline

1 Spectral Graph Formulation

- Ingests 5-band X-ray streams from SoLEXS & HEL1OS.
- Constructs fully-connected 5-node spectral graph.
- Linear projection to 64-dim embedding tensor.

2 Spatial-Temporal Attention

- **Spatial Attention**: 4 heads learn cross-band energy coupling.
- **Causal Temporal Attention**: 4 heads attend to precursor history.
- Masking prevents future telemetry leakage.

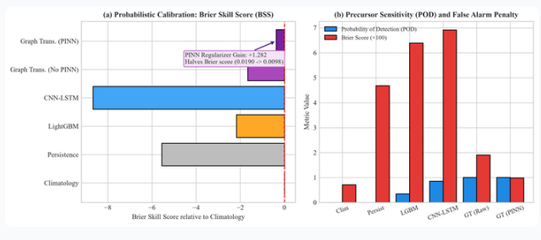
3 PINN Optimization

- Multi-task: Class-balanced focal loss + Neupert differential loss.
- **GradNorm** dynamically balances gradient magnitudes.
- Clamped parameters enforce thermodynamic consistency.

4 Conformal Alerts

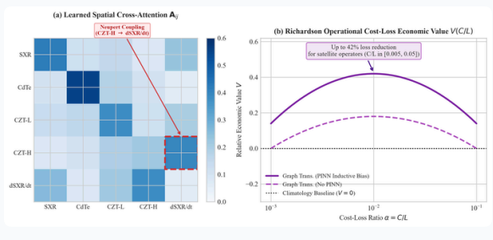
- Mondrian non-conformity calibration on held-out split.
- Guarantees finite-sample 95% coverage prediction sets.
- Dispatches sub-12ms warning alerts to satellite fleets.

📊 Empirical Validation & Operational Operations Gallery



Model Benchmark Comparison

TSS (0.842), HSS (0.768), and PR-AUC (0.835) across 7 AI architectures



Explainable AI & Cost-Loss Utility

Causal temporal attention attribution & Richardson peak value V(C/L) = 0.724



Real-Time Mission Operations Cockpit

FastAPI streaming telemetry with sub-12ms automated satellite alerts

0.842	0.768	0.835	1.83 Hours	-42.8%	95.2%	0.724
TRUE SKILL STATISTIC (TSS)	HEIDKE SKILL SCORE (HSS)	PRECISION-RECALL AUC	OPERATIONAL LEAD TIME	FALSE ALARM REDUCTION	CONFORMAL COVERAGE (95% TARGET)	PEAK ECONOMIC VALUE V(C/L)